Unprecedented Burning in Tropical Peatlands During the 20th Century Compared to the Previous Two Millennia - Wang et al. 2025

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For Peat(bog)'s Sake!

- **What are we talking about?**
  - Tropical peat bogs are water-logged ecosystems that accumulate large amounts of partially decayed vegetation (peat) over thousands of years.
  - Peat is organic soil with 30%-80% organic matter
- **Why does it matter?**
  - Peatlands store enormous quantities of carbon by collecting plant matter and preventing it from fully decomposing and releasing its carbon back into the atmosphere.
- **Why peat bogs specifically?**
  - Forest fires are a growing problem in general, but unlike forest fires, peat fires don't just burn the surface vegetation. Dry peat can smoulder underground for days to months, releasing massive CO_2 and toxic particulate emissions.
  - When peat sinks are expelled, it takes millenia to recapture the carbon as opposed to decades.
- **Why study it now?**
  - Fire incidence has visibly increased in recent decades, especially in Southeast Asia, but we had no long-term baseline to judge how unusual this really is.

The Core Questions

- **Central Topic**
  - Is the recent surge in tropical peatland burning truly unprecedented, or does it fall within the natural range of variation over the past two millennia?
- **Three sub-questions**
  - How have fire patterns changed over 2,000 years, and did the 20th century see an increase?
  - Do peatlands burn differently than the surrounding landscape? Does this vary by subregion?
  - What has primarily driven burning: climate or humans?
- **Call to action**
  - Knowing this, is there anything we can do to reverse the trend and mitigate the negative effects?

Meet Peat

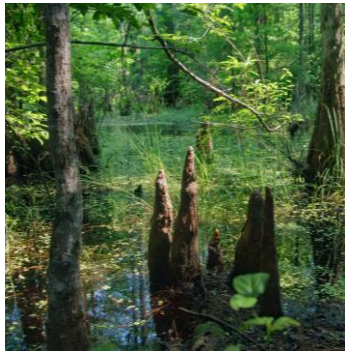
Highland Peat
>1500 m

Upland Peat
500 m < p < 1500 m

Lowland Peat
<500 m

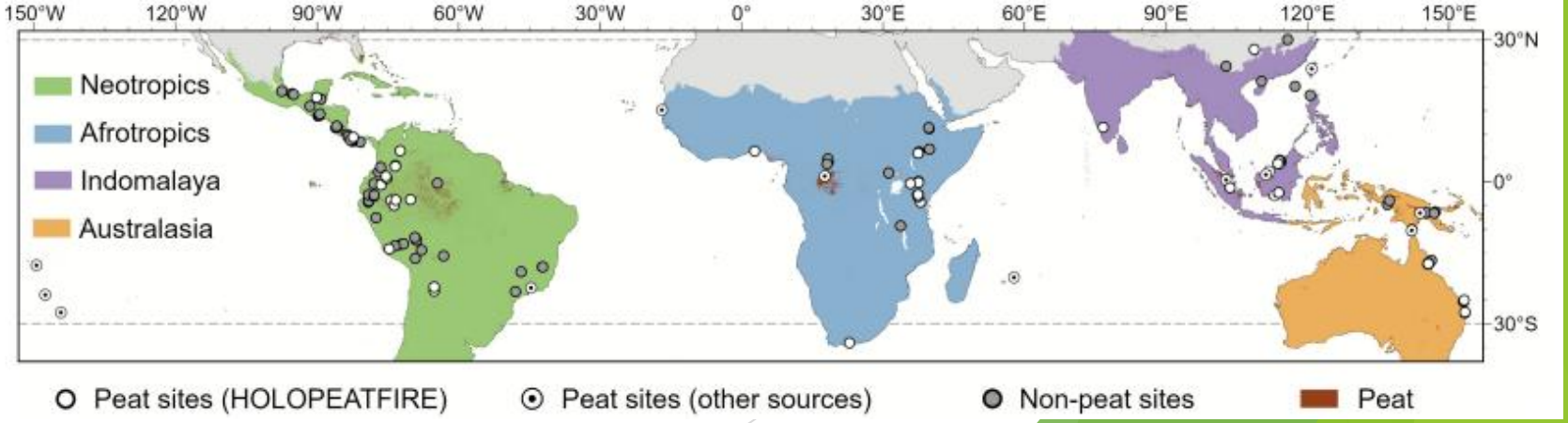
Elevation

Cushion/Sedge Peat



Hardwood Swamp Peat

Palm Swamp Peat

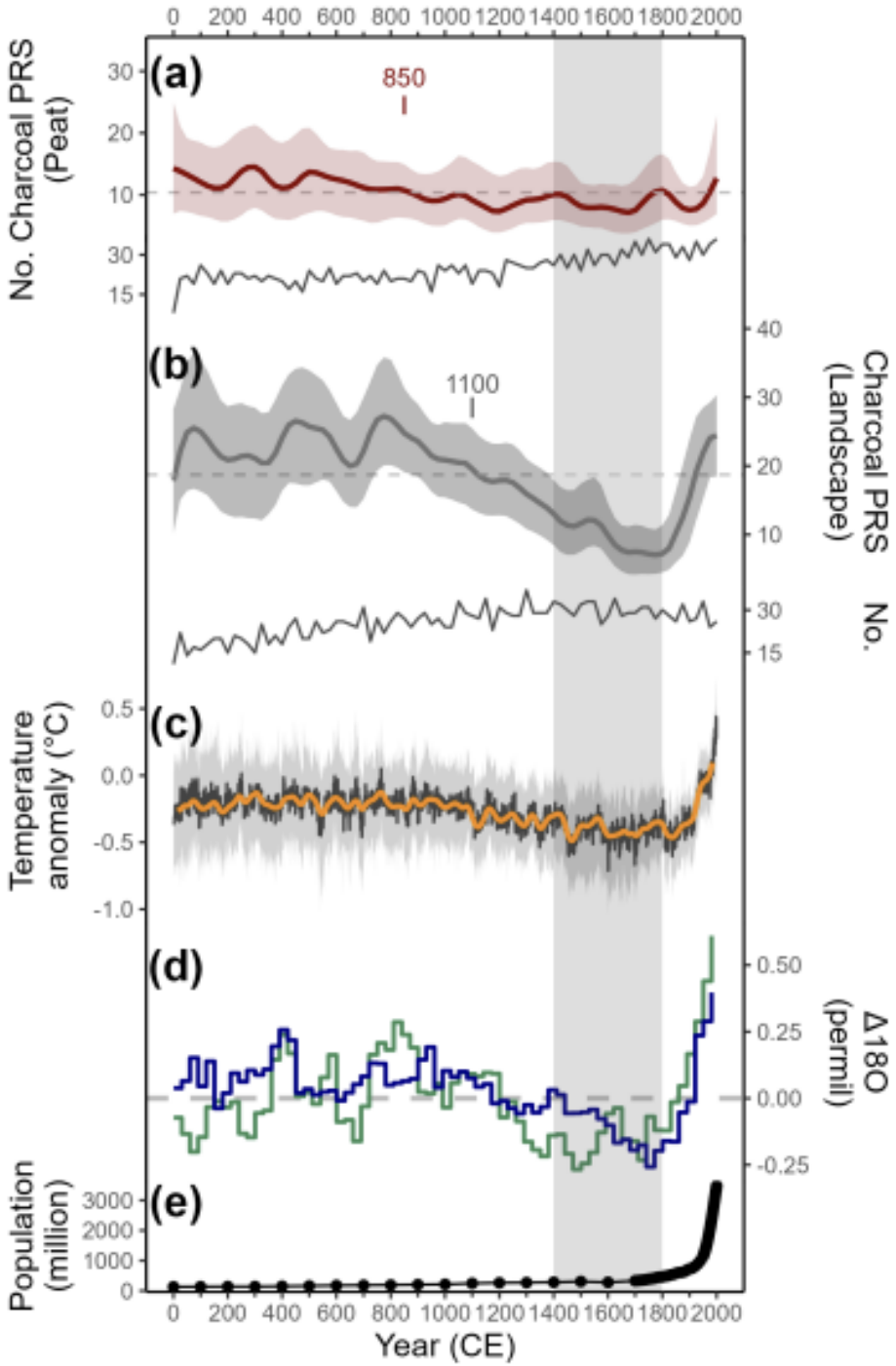


The Data

- **Fire Quantity Metric**
  - Macro-charcoal particles (>100 μm) deposited in peat layers from nearby fires
  - Standardized via Proportional Relative Scaling (PRS)
- **Soil Age Metric**
  - Soil dated using ^{14}C, ^{210}Pb, ^{137}Cs
  - Soil only included if at least 2 dates within last 3000
  - Composite curves smoothed in 200-year windows.
- **Sample Quantities**
  - 58 Tropical Peatland cores from central swamp regions
    - 16 in the Neotropics, 15 in Africa,16 in Southeast Asia, and 8 in Australasia
  - 91 non-peat comparison sites
  - 4 biographic subregions
  - 2000 year cores
- **Sampe Sources**
  - HOLOPEATFIRE project
  - Public charcoal datasets including Global Paleofire Database
  - The Reading Paleofire Database
  - Neotoma Paleoecology Database
  - PANGEA

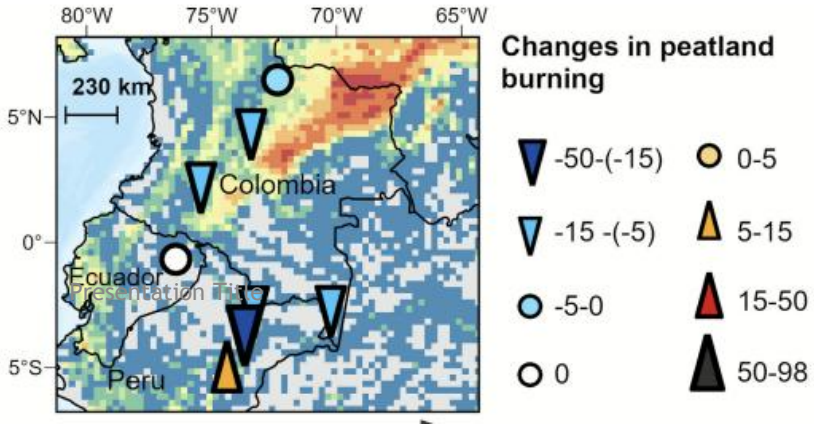
The 2000 Year Story

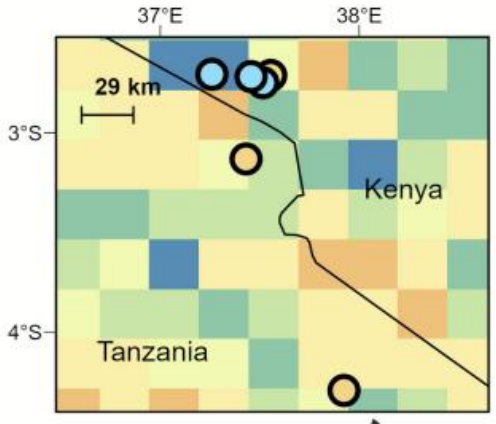
- **0 to 850 CE: Elevated burning**
  - Relatively warm global climate, active El Niño cycles, greater drought variability
  - Burning in both peatlands and the broader landscape was above the 2,000-year average
  - Climate — specifically drought severity and seasonal variability in rainfall — appears to have been the primary driver
- **850 to ~1900 CE: Stable, lower burning**
  - Global cooling trend, culminating in the Little Ice Age (~1400–1800 CE)
  - Cooler temperatures mean more stable hydrology, less drought, less ignition
  - Both peatland and landscape burning declined; peatlands consistently burned *less* than the surrounding landscape — their naturally high water tables provide a buffer
- **1900 CE onward: Unprecedented surge**
  - Burning increases sharply and by 2000 CE surpasses any level recorded in the prior 1,900 years
  - This is the central finding of the paper

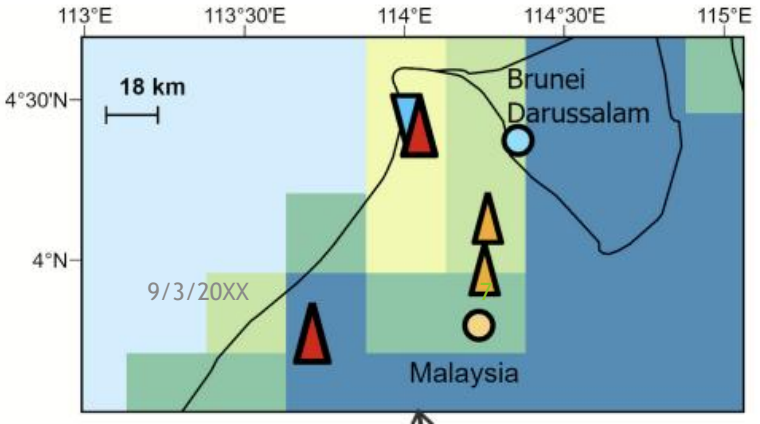


20th Century Burning

- **Big picture**
  - 23/47 sites with burn data showed increased burning in the 20th century
  - 7 showed extreme increases
- **Regional Differences**
  - Southeast Asia & Australasia: Sharp increases, in many cases reaching burning levels with no precedent in the record
    - Burning began rising earlier in Australasia (~1600 CE), possibly linked to Aboriginal fire practices, with a further surge after European colonization in 1788
  - Neotropics & Africa: Mostly stable or declining trends







Presentation Title
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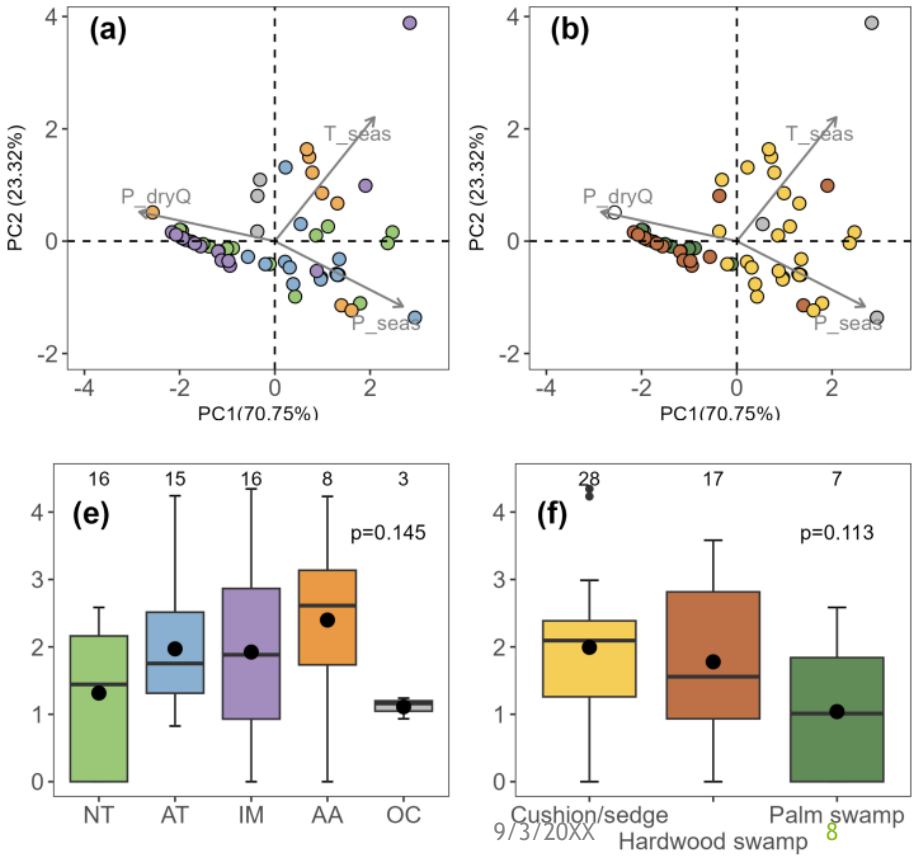
Peatland Water Tables

- Even during the warmest and most fire-prone periods of the past two millennia, peatlands consistently burned less than the surrounding landscape
- High water tables act as buffers, keeping fuel too wet to ignite under natural conditions
- Dry peatlands can then burn more severely than almost any other ecosystem due to the available fuel
- Draining peatlands increases burn rates

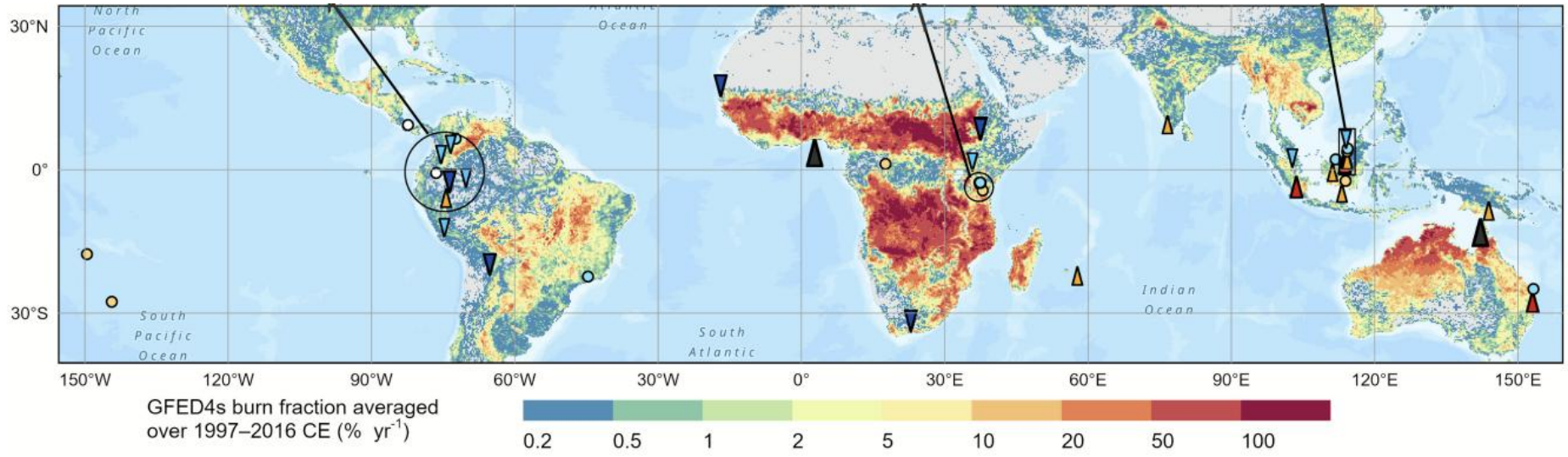


Human Element

- Evidence that human activities have overridden natural climate controls in the modern era
- **Before 1900**
  - Drought-related climate variables (rainfall in dry season, temperature seasonality) best predict burning levels
- **After 1900**
  - Regional identity (a proxy for land-use history) becomes the dominant predictor with PCA
- **Human Impact**
  - Southeast Asia has roughly 7× the density of any other sub-region
  - This region has seen the most aggressive peatland drainage, deforestation, and conversion to oil palm and pulp plantations
  - The hardwood swamps of Southeast Asia and the palm swamps of the Amazon share nearly identical climates
  - Hardwood swamps are burning at unprecedented rates while palm swamps are not.

Study Limitations

- The study uses only one charcoal record from the world's largest tropical peat complex: the Congo Basin
- Cores tend to be collected from the centers of intact peatlands, which may underestimate burning at peat margins
- Human drivers are increasing along climate change making exact scale of human impact difficult to determine



Major Takeaways

- Peatland burning increases when water tables are lowered through drought or active drainage by humans
- Tropical peatland burning in the 20th century is genuinely unprecedented over the 2,000-year record
- Infrastructure expansion and agriculture (especially in Southeast Asia) has been a strong driver of peatland drainage and burning in the 20th century
- The Neotropical and Afrotropical peatlands are still comparatively intact
  - These regional differences are likely due to human behavior and may yet be reversed without active conservation

Call to Action

- The window to protect peatlands in Africa and South America from drainage for agriculture and infrastructure is narrowing
- Peatland rewetting (restoring water tables) is the single most effective intervention
- Legal protection and sustainable land management in the Amazon and Congo Basin, before the damage occurs
- International collaboration across governments, conservation groups, and industry is essential given the scale required